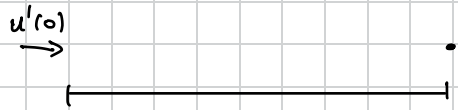
VERIFICATION

PRESSURE

$$\begin{cases} u''(x) + k^2 u(x) = 0 \\ u'(0) = -\rho \omega^2 A \\ u'(1) = 0 \end{cases} \qquad k^2 = \frac{\omega^2}{c^2} \qquad x \in (0,1)$$



$A = 0.001 \qquad \rho = 1{,}225 \ Kg/m^3 \qquad c = 340 \ m/s$

$$\omega = \begin{cases} 100 \ 2\pi \ Hz \\ 500 \ 2\pi \ Hz \end{cases}$$

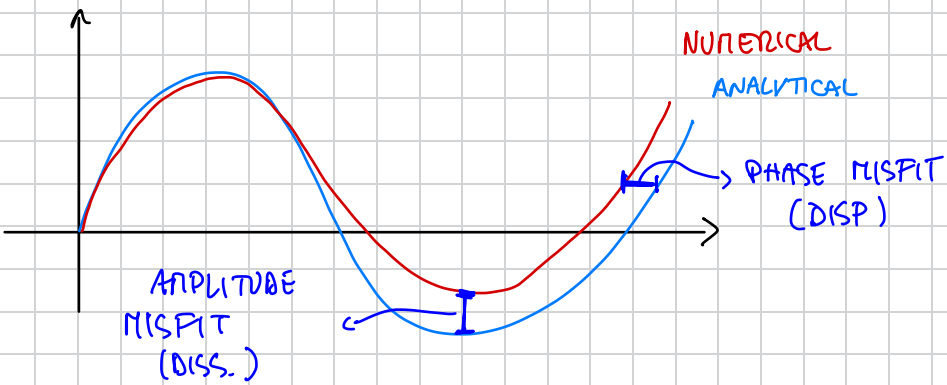
- COMPUTE ω_i when $u'(0) = 0$

1) $A \underline{u} = k_h^2 M \underline{u}$

2) $\omega_h^2 = k_h^2 \cdot c^2 \qquad \omega_h \approx \omega$

$$\omega_n = \frac{n\pi}{1} = n\pi$$

DISSIPATION AND DISPERSION ANALYSIS



$$(P)\begin{cases} \dfrac{\partial^2 u}{\partial t^2} - c^2 \dfrac{\partial^2 u}{\partial x^2} = 0 & x \in \mathbb{R} \\ u(x,0) = u_0(x) \\ \dfrac{\partial u}{\partial t}(x,0) = v_0(x) \end{cases}$$

- NO EXTERNAL SOURCES $(f=0!)$
- INFINITE DOMAIN
- c^2 IS CONSTANT

$$\underline{u(x,t)} = e^{i(\omega t - Kx)} \qquad K = \frac{2\pi}{\lambda} \qquad \omega = \frac{2\pi}{T}$$

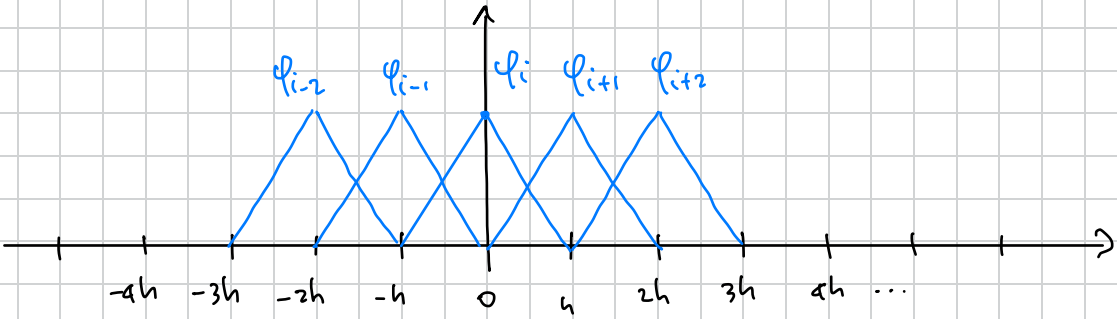
Substitute into the eq. we get

$$-\omega^2 u(x,t) + c^2 K^2 u(x,t) = 0$$

$$u(x,t)\left(k^2c^2-\omega^2\right)=0 \iff c^2=\frac{\omega^2}{k^2}$$

DISPERSION RELATION

DISCRETIZE (P) with FE



$$\begin{cases} M\ddot{\underline{u}} + c^2 A\underline{u} = \underline{0} & t>0 \\ \underline{u}(0)=\underline{u}_0 \\ \dot{\underline{u}}(0)=\underline{v}_0 \end{cases}$$

INFINITE DIM ODE SYSTEM !

look at the i-th equation !

$\underline{v} = \varphi_i(x)$

$$m_{i\,i-1}\,\ddot{u}_{i-1}(t) + m_{ii}\,\ddot{u}_i(t) + m_{i\,i+1}\,\ddot{u}_{i+1}(t)$$
$$= -c^2\left(a_{i\,i-1}\,u_{i-1}(t) + a_{ii}\,u_i(t) + a_{i\,i+1}\,u_{i+1}(t)\right)$$

$$\frac{h}{6}\ddot{u}_{i-1}(t) + \frac{2}{3}h\,\ddot{u}_i(t) + \frac{h}{6}\ddot{u}_{i+1}(t)$$

$$= -c^2\left(-\frac{1}{h}\underline{u_{i-1}(t)} + \frac{2}{h}u_i(t) - \frac{1}{h}u_{i+1}(t)\right)$$

$$u_i(t) = u_h(x_i, t) = e^{i(\omega t - kx_i)}$$

$$u_{i-1} = e^{i(\omega t - kx_{i-1})} = e^{i(\omega t - kx_i)}\,e^{ikh}$$

$$= u_i\,e^{ikh}$$

$$x_{i-1} = x_i - h$$

$$x_{i+1} = x_i + h$$

$$u_{i+1} = u_i\,e^{-ikh}$$

$$\left(\frac{h^2}{6}e^{ikh} + \frac{2}{3}h^2 + \frac{h^2}{6}e^{-ikh}\right)\ddot{u}_i$$

$$= -c^2\left(-\frac{1}{\cancel{h}}e^{ikh} + \frac{2}{\cancel{h}} - \frac{1}{\cancel{h}}e^{-ikh}\right)u_i$$

$$h^2\left(\frac{e^{ikh}}{6} + \frac{2}{3} + \frac{e^{-ikh}}{6}\right)\ddot{u}_i$$

$$= -c^2\left(-e^{ikh} + 2 - e^{-ikh}\right)u_i$$

$$\ddot{u}_i^k \simeq \frac{U_i^{k-1} - 2U_i^k + U_i^{k+1}}{\Delta t^2}$$

$$\parallel$$

$$\ddot{U}_i(t_k)$$

$$u_i^m = e^{-i(\omega t_m - k x_i)}$$

$$u_i^{m-1} = e^{i(\omega(t_m - \Delta t) - k x_i)} = u_i^m e^{-i\omega\Delta t}$$

$$U_i^{m+1} = u_i^m e^{i\omega\Delta t}$$

$$h^2\left(\frac{e^{ikh}}{6} + \frac{2}{3} + \frac{e^{-ikh}}{6}\right)\left(\frac{e^{-i\omega\Delta t} - 2 + e^{i\omega\Delta t}}{\Delta t^2}\right) u_i^m$$

$$= -c^2\left(-e^{-ikh} + 2 - e^{ikh}\right) u_i^m$$

$$u_i^m \underbrace{\left(\quad \circledast \quad \right)} = 0$$

$$e^{ikh} = \cos(kh) + i\,\mathrm{sen}(kh) \ldots$$

$$\frac{4}{\Delta t^2} \sin\left(\frac{\omega \Delta t}{2}\right) = 3 \frac{c^2}{h^2} \frac{2 - 2\cos(kh)}{2 + 2\cos(kh)}$$

DISCRETE DISPERSION RELATION

$\omega \Delta t \to 0$ $\qquad$ $kh \to 0$

$$\omega^2 \simeq c^2 k^2$$

$\omega \Delta t \to 0 \qquad \omega \nearrow \qquad \Delta t \ll 1$

$kh \to 0 \qquad k \nearrow \qquad h \ll 1$

$\Rightarrow$ FE method is accurate for large wave number or high frequency if the mesh size is small or time step is small.

- $u(x,t) = \cos(kx - \omega t) \qquad \omega^2 = c^2 k^2$
- PERIODIC B.C. $\qquad c^2 = 1$

SPECTRAL ELEMENT METHOD (HIGH-ORDER FEM)

1) Special basis function (ORTHOGONAL POLYNOMIALS)

2) Basis function are defined using suitable grid points (NOT EQUISPACED)

3) Appropriate quadrature formulas to compute integrals

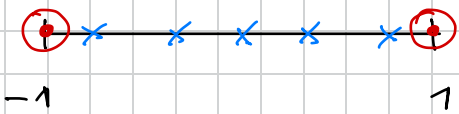
1) $I = [-1,1]$, define Legendre polynomials in I

$$\begin{cases} L_0 = 1 \qquad L_1 = x \\ L_{k+1} = \dfrac{2k+1}{k+1} x L_k - \dfrac{k}{k+1} L_{k-1} \qquad k = 1, 2 \dots \end{cases}$$

$L_k \in \mathbb{P}^k(I)$, $\{L_k\}_{k=0}^{\infty}$ are orthogonal in I

$$\int_{-1}^{1} L_m(x) L_n(x)\, dx = \begin{cases} 0 & n \neq m \\ \left(m + \frac{1}{2}\right)^{-1} & n = m \end{cases}$$

2) $I = [-1,1]$



$$\begin{cases} x_0 = -1 \\ x_n = 1 \\ \{x_i\}_{i=1}^{n-1} \text{ s.t. } L_n'(x_i) = 0 \end{cases}$$

ZEROS OF L_n' all inside $(-1,1)$

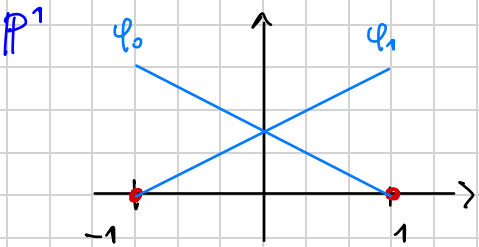
N+1 grid points inside $[-1,1]$

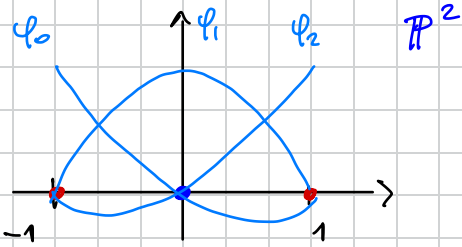
→ GAUSS-LEGENDRE-LOBATTO (GLL) POINTS

BASIS FUNCTION $\varphi_i(x) \in \mathbb{P}^n$

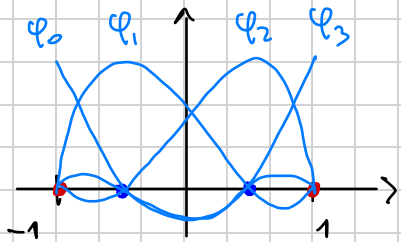
$$\varphi_i(x) = -\frac{1}{n(n+1)} \frac{(1-x^2)}{(x-x_i)} \frac{L_n'(x)}{L_n(x_i)} \qquad i = 0, \dots, n$$

$$\varphi_i(x_j) = \delta_{ij} \qquad \begin{cases} 1 & i = j \\ 0 & \text{otherwise} \end{cases}$$

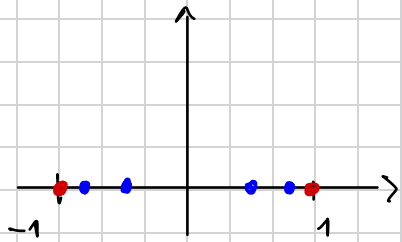




$\mathbb{P}^3$



$\mathbb{P}^4$



3) GLL - quadrature rule

$$\int_{-1}^{1} f(x)\,dx \simeq \sum_{i=0}^{n} \alpha_i f(x_i)$$

x_i are the GLL nodes

$$\alpha_i = \frac{2}{i(i+1)} \frac{1}{L_n^2(x_i)} > 0$$

GLL is exact up to polynomials of order $2n-1$.

FE FORM FOR 1D WAVE EQ. (NEUMANN COND)

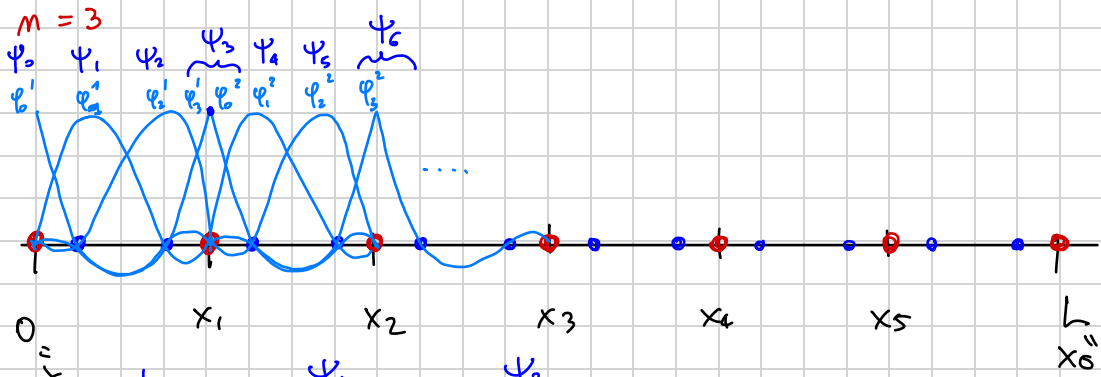
For any $t \in (0,T]$ find $u_h = u_h(t) \in V_h^n$ s.t.

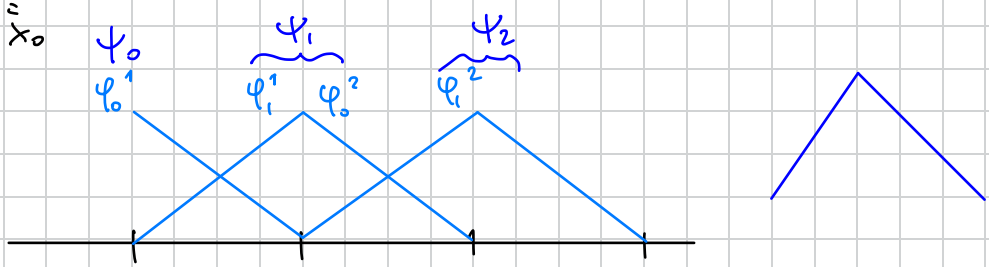
$$\int_0^L \rho \frac{\partial u_h}{\partial t^2} v_h \, dx + \int_0^L \mu \frac{\partial u_h}{\partial x} \frac{\partial v}{\partial x} \, dx = \int_0^L f v_h \, dx$$

$\forall v_h \in V_h$

+ INITIAL COND

$$V_h^n = \left\{ v \in C^0(0,L) : v\big|_{K_j} \in \mathbb{P}^n(K_j) \ \forall j \right\}$$





$$u_h(x,t) = \sum_{j=0}^{M} u_j(t)\, \psi_j(x)$$

$$M = ?$$

$$M = \overbrace{(N_h+1)}^{\text{grid nodes}} + \underset{\text{el.}}{N_h}\,(n-1)$$

$$= N_h + 1 + n N_h - N_h = n N_h + 1$$

$n = 1 \rightarrow M = N_h + 1$ (grid points)

$n = 2 \rightarrow M = 2N_h + 1 = (N_h + 1) + N_h$

$$M \ddot{\underline{U}} + A \underline{U} = \underline{F}(t)$$

$\downarrow$ M is diagonal $\begin{pmatrix} \ddots & 0 \\ 0 & \ddots \end{pmatrix}$

EXPLICIT SCHEMES

$$M \underline{U}^{k+1} = f(\underline{U}^k, \underline{U}^{k-1}, \dots)$$

$$M_{ij} = \int_0^L \rho \, \psi_i \psi_j \, dx$$

$$= \sum_{m=1}^{N_h} \int_{K_m} \rho \, \psi_i \psi_j \, dx$$

$$= \sum_{m=1}^{N_h} \int_{-1}^{1} \rho(\xi) \hat{\psi}_j(\xi) \hat{\psi}_i(\xi) \frac{h}{2} \, d\xi$$

$$\simeq \sum_{m=1}^{N_h} \sum_{q=1}^{n+1} \rho(\xi_q) \underbrace{\hat{\psi}_j(x_q)}_{\delta_{jq}} \underbrace{\hat{\psi}_i(x_q)}_{\delta_{iq}} \frac{h}{2} \underbrace{w_q}_{\text{weights}}$$

GLL points

$$= \begin{cases} \sum_{m=1}^{N_h} \rho(\xi_i) \frac{h}{2} w_i & \text{if } i = j \end{cases}$$

$\begin{cases} \\ 0 & \text{otherwise} \end{cases}$